# ADAM D. RUSHDY

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## EDUCATION

**Purdue University** - B.S. Aeronautical & Astronautical Engineering, Honors *Expected Graduation: May 2028*

**GPA:** 3.82/4.00

**Activities**: Purdue Aerial Robotics Team, Teaching Assistant, All American Marching Band (Assistant Section Leader)

## EXPERIENCE

**Engineering Intern** - *Xchanger* *Hopkins, MN | May 2026 - Present*

- Tested pressure-loss performance of air-to-air and gas-to-gas heat exchangers across 9 flow conditions using the Log-Tchebycheff traversal method (18 points/measurement), improving data-acquisition accuracy by 10% and identifying a 2-3x underestimation in predicted pressure-drop, enabling model recalibration.
- Generated performance curves for 77% of air-air units across multiple configurations by combining empirical airflow and differential pressure data.
- Configured experimental airflow test sections with minimum 8 duct diameters upstream and 3 downstream of sampling locations, minimizing turbulence effects and improving measurement reliability.

**Teaching Assistant** - *Honors Intro to Engineering (ENGR 16X)* *West Lafayette, IN | Aug 2025 - Present*

- Support 60+ first-year students with coursework and grading including 6 design projects spanning Python modeling, MATLAB simulation, and Raspberry Pi embedded systems.

**Intern** - *Seminar for Top Engineering Prospects (STEP)* *West Lafayette, IN | Jun - Aug 2025*

- Developed and delivered a modular Arduino programming curriculum to 320 prospective engineers, built functions and tutorials to teach students basic programming skills.

**Teaching Assistant** - *Intro to C Programming* *West Lafayette, IN | Jan - May 2025*

- Led 30+ students across 12 labs covering core C concepts, held office hours, and graded coding assignments.

## PROJECTS

**R&D Airframe Member** - *Purdue Aerial Robotics Team (PART)* *Sep 2025 - Present*

- Designed and manufactured 9 wing structure and tail assembly components for a team eVTOL aircraft focused on modularity and easy assembly with SolidWorks.
- Developed MATLAB structural analysis to evaluate rib spacing trade-offs between panel buckling and structural weight, informing airframe design decisions.
- Developed CAM toolpaths for CNC machining of 2 precision carbon-fiber layup wing molds within a 2 hour manufacturing constraint using Fusion 360.
- Fabricated 2 composite wing assemblies using carbon fiber layup over laser-cut internal structures.

**9-State Inertial Navigation System** *June - July 2026*

- Developed an INS with Python using ESP-32, IMU, GPS, and magnetometer fusing 9 sensor streams via quaternion-based 6-DOF complementary and Kalman filtering at 40 Hz for real-time state estimation.
- Reduced position RMSE from 13 m to 4.9 m (62% improvement) by diagnosing attenuation of high-dynamics events and re-tuning Kalman noise covariances, benchmarked against ArduPilot EKF logs to achieve <5 m error.

**Fixed-Wing Surveillance UAV - Design, Analysis & Fabrication** *Mar - June 2026*

- Designed, analyzed, and fabricated a 1.3 m, 1.8 kg fixed-wing UAV with raspberry pi camera integration. Selected airfoils in XFOIL, sized wing geometry for 25 m/s cruise, and modeled the full assembly in SolidWorks.
- Performed aerodynamic, stability, and flight dynamics analysis in XFLR5, iterating on CG, tail moment arm, and wing geometry to achieve a 10% static margin.
- Designed wing structure for a 6g limit load factor, used finite element analysis to confirm structural integrity with a maximum tip deflection of 12.1 mm. Fabricated using carbon-fiber skin layups over 3D-printed internal ribs and spars. Measured 20% mass difference (1.5 kg predicted) attributed to additive manufacturing density uncertainty.
- Implemented ArduPilot flight controller for fly-by-wire and data logging. Completed maiden flight capturing airspeed, load factor, battery performance, and estimated position, across 150 second flight, validating ~25 m/s cruise speed and maximum 3g load against design predictions.

## SKILLS

**CAD / Analysis:** SolidWorks, Fusion 360, NX, Teamcenter, XFLR5, XFOIL

**Programming:** MATLAB/Simulink, Python, C/C++, Java, Arduino, Git/GitHub

**Fabrication:** Carbon-fiber composite layup, 3D printing (FDM), CNC milling

**Additional:** Eagle Scout, BSA Troop 479 (2017–2023) — 150+ volunteer hours with team leadership experience